# Machine Learning and Data Mining (ST443)

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LSE

# Content: linear regression (Ch 3 @ ISLP, Ch 3.2 @ ESL)

- probability recap,
- objective, solution, projection interpretation,
- testing, confidence interval,
- measure of fit,
- linear regression tweaks, towards non-linear methods.

# Probability recap: continuous distributions on $\mathbb{R}^d$

Representation of $\mathbb{P}$: pdf (probability density function) $f$

1. non-negative:

$$f(\mathbf{x}) \geq 0 \text{ for all } \mathbf{x} \in \mathbb{R}^d.$$

2. normalized:

$$\int_{\mathbb{R}^d} f(\mathbf{x}) \mathrm{d}\mathbf{x} = 1.$$

Meaning: $X \sim \mathbb{P}$, with pdf $f_X$, for 'any' set $B \subseteq \mathbb{R}^d$

$$\mathbb{P}(X \in B) = \int_B f_X(\mathbf{x}) \mathrm{d}\mathbf{x}.$$

# Continuous distribution – continued

Alternative representations:

- Cdf (cumulatived density function): $X \sim \mathbb{P}$,

$$F(\mathbf{z}) = \mathbb{P}(X < \mathbf{z}) = \mathbb{P}(X \in \underbrace{\times_{j=1}^{d}(-\infty, z_j)}_{=B\text{: interval}}), \quad \mathbf{z} = (z_j)_{j=1}^{d} \in \mathbb{R}^d.$$

- Characteristic function ($i = \sqrt{-1}$):

$$\phi_X(\mathbf{t}) = \mathbb{E}\left[e^{i\langle \mathbf{t}, X\rangle}\right], \quad \mathbf{t} \in \mathbb{R}^d.$$

**At the moment**

pdf will the job.

# Usage of pdf: $d = 1$

- Expectation, moments:

$$\mathbb{E}[X] = \int_{\mathbb{R}} x f(x) \mathrm{d}x, \quad \mathbb{E}[X^n] = \int_{\mathbb{R}} x^n f(x) \mathrm{d}x \quad (n \in \mathbb{Z}^+).$$

- Covariance of $(X, Y) \in \mathbb{R} \times \mathbb{R}$:

$$\mathrm{cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

- Variance, standard deviation:

$$\mathrm{var}(X) = \mathrm{cov}(X, X), \qquad \sigma(X) = \sqrt{\mathrm{var}(X)}.$$

- Correlation:

$$\mathrm{corr}(X, Y) = \frac{\mathrm{cov}(X, Y)}{\sigma(X)\sigma(Y)}.$$

# Properties ($d = 1$)

- Covariance, with constants $a, b \in \mathbb{R}$:

$$\begin{aligned} \operatorname{cov}(X, a) &= 0, \\ \operatorname{cov}(X, Y) &= \operatorname{cov}(Y, X), \\ \operatorname{cov}(aX, bY) &= ab \operatorname{cov}(X, Y), \\ \operatorname{cov}(X + a, Y + b) &= \operatorname{cov}(X, Y). \end{aligned}$$

- Properties of variance:

$$\operatorname{var}(X) \geq 0, \qquad \operatorname{var}(X) = 0 \Leftrightarrow X \equiv c \text{ (a.s.)}.$$

- Correlation (Pearson's correlation coefficient), with constant $a, b \in \mathbb{R}$:

$$\operatorname{corr}(X, Y) \in [-1, 1], \quad |\operatorname{corr}(X, Y)| = 1 \Leftrightarrow X = aY + b.$$

# Usage of pdf: $d \geq 1$

- Expectation:

$$\mathbb{E}[X] = \int_{\mathbb{R}^d} \mathbf{x} f(\mathbf{x}) \mathrm{d}\mathbf{x}.$$

- Covariance of $(X, Y) \in \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$:

$$\mathrm{cov}(X, Y) = \mathbb{E}\left[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])^\top\right] = \mathbb{E}\left[XY^\top\right] - \mathbb{E}[X](\mathbb{E}[Y])^\top.$$

- Variance:

$$\mathrm{var}(X) = \mathrm{cov}(X, X).$$

# Properties ($d \geq 1$)

- Mean ($\mathbf{A} \in \mathbb{R}^{d_1 \times d_2}$, $\mathbf{b} \in \mathbb{R}^{d_2}$, $X \in \mathbb{R}^{d_2}$):

$$\mathbb{E}[\mathbf{A}X + \mathbf{b}] = \mathbf{A}\mathbb{E}[X] + \mathbf{b} \text{ (under affine transformation)}.$$

- Variance:

$$\begin{aligned} \operatorname{var}(X) &= [\operatorname{var}(X)]^\top \text{ (symmetric)}, \\ \operatorname{var}(X) &: \text{ positive semidefinite, i.e. } \mathbf{v}^\top \operatorname{var}(X)\mathbf{v} \geq 0 \text{ for all } \mathbf{v}, \\ \operatorname{var}(\mathbf{A}X + \mathbf{b}) &= \mathbf{A} \operatorname{var}(X)\mathbf{A}^\top \text{ (under affine transformation)}. \end{aligned}$$

- Covariance:

$$\begin{aligned} \operatorname{cov}(X, Y) &= [\operatorname{cov}(Y, X)]^\top, \\ \operatorname{cov}(\mathbf{A}X + \mathbf{b}, \mathbf{C}Y + \mathbf{d}) &= \mathbf{A} \operatorname{cov}(X, Y)\mathbf{C}^\top. \end{aligned}$$

# Celebrities in the area: normal distribution ($d \geq 1$)

- $\mathbb{R}^d \ni X \sim \mathcal{N}(\mathbf{m}, \mathbf{\Sigma})$ has pdf

$$f_X(\mathbf{x}) = (2\pi)^{-d/2}[\det(\mathbf{\Sigma})]^{-1/2} e^{-\frac{1}{2}(\mathbf{x}-\mathbf{m})^\top \mathbf{\Sigma}^{-1}(\mathbf{x}-\mathbf{m})}.$$

- Expectation, covariance:

$$\mathbb{E}[X] = \mathbf{m}, \qquad \mathrm{var}(X) = \mathbf{\Sigma}$$

  completely characterizes the distribution.

- Under affine transformation:

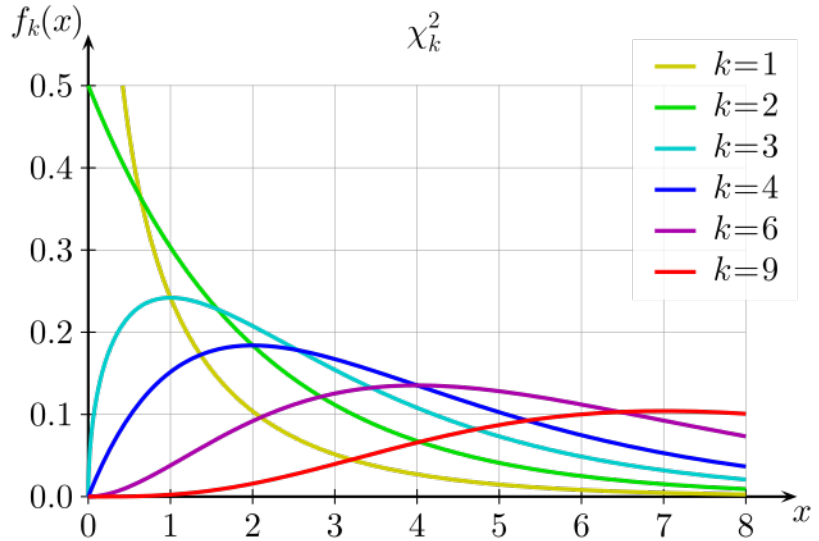
$$\mathbf{A}X + \mathbf{b} \sim \mathcal{N}\left(\mathbf{Am} + \mathbf{b}, \mathbf{A\Sigma A}^\top\right).$$

# Celebrities+: $\chi^2$ distribution ($d = 1$)

- Definition ($k \in \mathbb{Z}^+$ degrees of freedom):

$$\chi^2_k = \sum_{i=1}^{k} X_i^2,$$

where $X_1, \ldots, X_k$ are independent and $\mathcal{N}(0, 1)$-distributed.

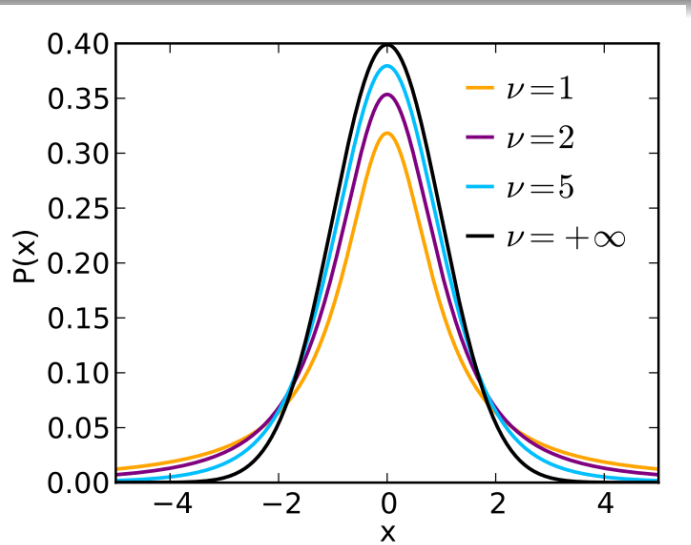



- Pdf:

$$f_k(x) = \frac{1}{2^{\frac{k}{2}}\Gamma\left(\frac{k}{2}\right)} x^{\frac{k}{2}-1} e^{-\frac{x}{2}},\ x \in (0, \infty), \quad \Gamma(z) = \int_{\mathbb{R}_{\geq 0}} t^{z-1} e^{-t} \mathrm{d}t\ (z > 0).$$

- Mean, variance: $\mathbb{E}[X] = k$, $\mathrm{var}[X] = 2k$.

# Celebrities+: t-distribution ($d = 1$)



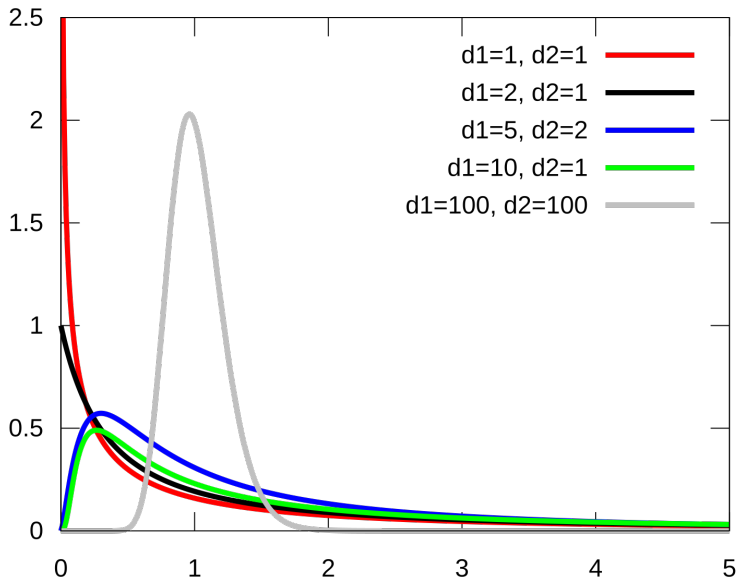

- Pdf ($\nu$ = degrees of freedom $> 0$):

$$f_\nu(x) = \frac{1}{\sqrt{\nu}B\left(\frac{1}{2},\frac{\nu}{2}\right)} \frac{1}{\left(1+\frac{x^2}{\nu}\right)^{\frac{\nu+1}{2}}} \quad (x \in \mathbb{R}),$$

$$B(z_1,z_2) = \int_0^1 t^{z_1-1}(1-t)^{z_2-1}\mathrm{d}t = \frac{\Gamma(z_1)\Gamma(z_2)}{\Gamma(z_1+z_2)}, \ (z_1,z_2>0).$$

- Mean: 0 for $\nu > 1$; undefined otherwise.
- Variance: $\frac{\nu}{\nu-2}$ for $\nu > 2$; $\infty$ for $\nu \in (1,2]$; undefined otherwise.
- Limits: $\nu \to 1$: Cauchy distribution; $\nu \to \infty$: $\mathcal{N}(0,1)$.

# Celebrities+: $F$ distribution ($d = 1$)




- Def ($d_1, d_2 \in \mathbb{Z}^+$ degrees of freedom):

$$X = \frac{U_1/d_1}{U_2/d_2},$$

where $U_1 \sim \chi^2_{d_1}$ and $U_2 \sim \chi^2_{d_2}$ independent.

# Celebrities+: $F$ distribution ($d = 1$) – continued

- Pdf:

$$f_{d_1,d_2}(x) = \frac{\sqrt{\frac{(d_1 x)^{d_1}(d_2)^{d_2}}{(d_1 x + d_2)^{d_1+d_2}}}}{xB\left(\frac{d_1}{2}, \frac{d_2}{2}\right)}, \quad x \in (0, \infty) \text{ for } d_1 = 1, \text{ else } x \in [0, \infty).$$

- Mean and variance:

$$\mathbb{E}[X] = \frac{d_2}{d_2 - 2} \text{ for } d_2 > 2, \quad \text{var}(X) = \frac{2d_2^2(d_1 + d_2 - 2)}{d_1(d_2 - 2)^2(d_2 - 4)} \text{ for } d_2 > 4.$$

# Linear regression model

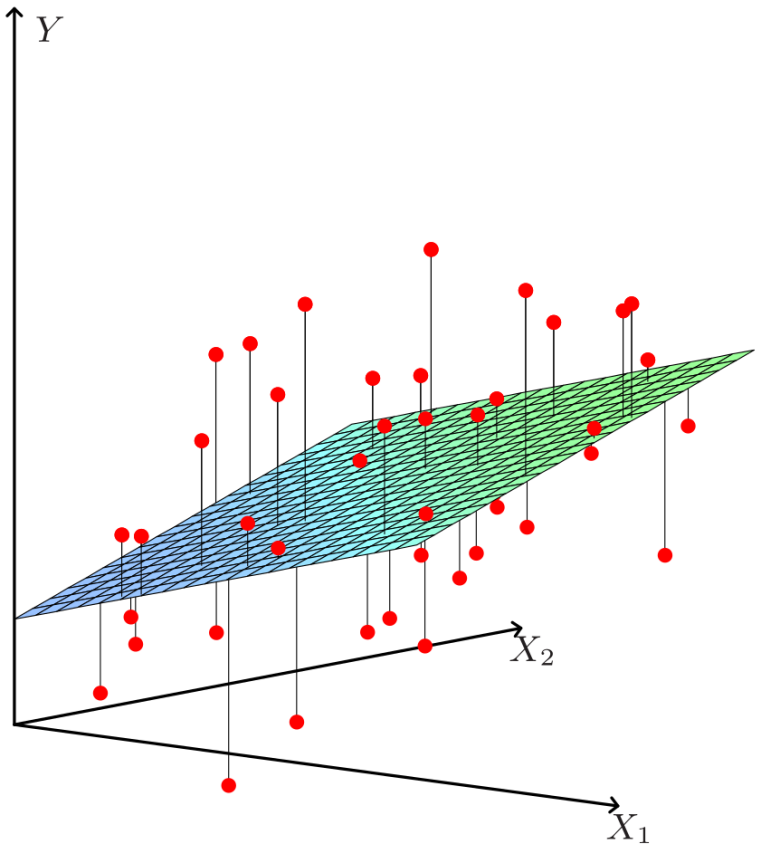
- Model:

$$Y = f(X) + \epsilon,$$
$$f(X) = f_\beta(X) = \beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p.$$

  with $Y \in \mathbb{R}$, $X = (X_i)_{i=1}^p \in \mathbb{R}^p$, $\beta = (\beta_i)_{i=0}^p \in \mathbb{R}^{p+1}$, $\epsilon \sim \mathcal{N}(0, \sigma^2)$.
- Simple and interpretable.
- Various generalizations (covered later).

# Towards learning: objective function

- Given: $\{(\mathbf{x}_i, y_i)\}_{i=1}^n \subset \mathbb{R}^p \times \mathbb{R}$ i.i.d. realizations; $\mathbf{x}_i = (x_{i,j})_{j=1}^p$.
- Method: least squares, minimizing the residual sum of squares; demo ($p = 2$):



# Objective function

Our goal:

$$\mathrm{RSS}(\beta) = \sum_{i\in[n]} [y_i - f_\beta(\mathbf{x}_i)]^2 = \sum_{i\in[n]} \left[ y_i - \beta_0 - \sum_{j\in[p]} x_{i,j}\beta_j \right]^2$$
$$= \|\mathbf{y} - \mathbf{X}\beta\|_2^2 \to \min_{\beta\in\mathbb{R}^{p+1}},$$

where $\mathbf{y} = (y_i)_{i=1}^n \in \mathbb{R}^n$, $\mathbf{X} = \left(\tilde{\mathbf{x}}_i^\top\right)_{i=1}^n \in \mathbb{R}^{n\times(p+1)}$, $\tilde{\mathbf{x}}_i = \begin{pmatrix} 1 \\ \mathbf{x}_i \end{pmatrix} \in \mathbb{R}^{p+1}$.

Note: residual standard error (normalized RSS):

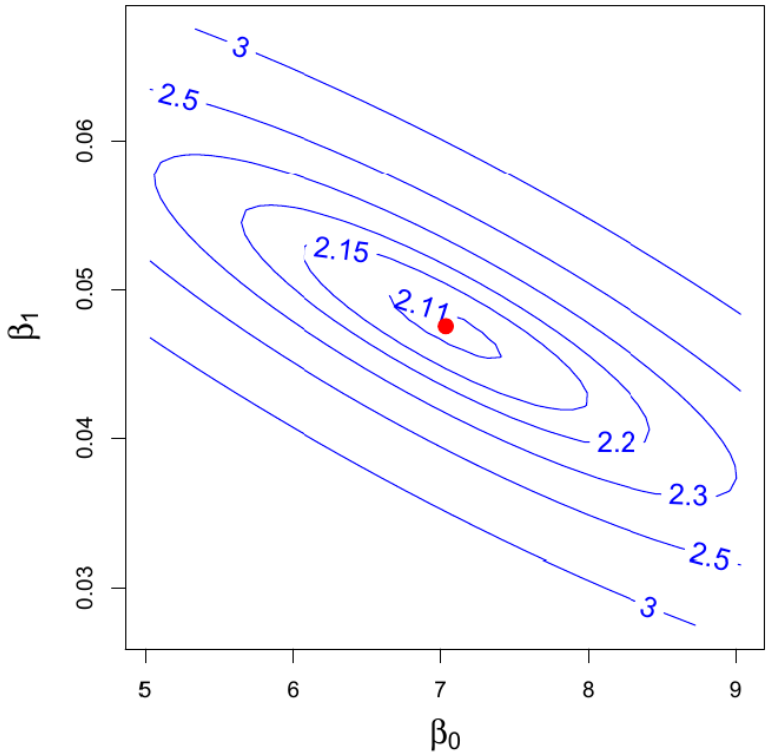
$$\mathrm{RSE} = \sqrt{\frac{1}{n-p-1}\mathrm{RSS}}.$$

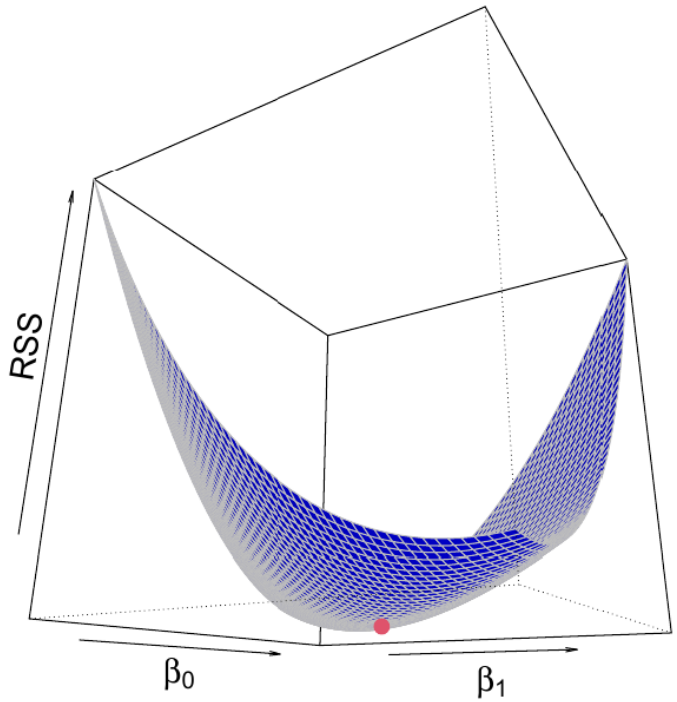
# Optimization: visual illustration

With the aim

$$\text{sales} \approx \beta_0 + \beta_1 \times \text{TV},$$

the objective



# Questions

In the example of

$$\text{sales} \approx \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{radio} + \beta_3 \times \text{newspaper},$$

natural questions:

1. Is there a relation between sales and advertising? ($H_0$: $\boldsymbol{\beta} = \mathbf{0}$ vs $H_1 : \boldsymbol{\beta} \neq \mathbf{0}$) $\Rightarrow$ hypothesis testing, F statistic.
2. How strong is the relation? (normalized RSS, percentage of variability in the response explained by the predictor) $\Rightarrow$ $\mathrm{RSE}$, $\mathrm{R}^2$.
3. Which media are associated with sales? ($H_0 : \beta_j = 0$ vs $H_1 : \beta_j \neq 0$) $\Rightarrow$ hypothesis testing, Z-score.
4. How large is the association between each medium and the sales? $\Rightarrow$ confidence interval for $\beta_j$ (or confidence set for $\boldsymbol{\beta}$).
5. Would non-linearity help? $\Rightarrow$ polynomial regression, feature map.

# Solution

$$\min_{\beta \in \mathbb{R}^{p+1}} \mathrm{RSS}(\beta) := \|\mathbf{y} - \mathbf{X}\beta\|_2^2 \overset{(a)}{=} (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta)$$

$$\overset{(b)}{=} \mathbf{y}^\top \mathbf{y} - 2\mathbf{y}^\top \mathbf{X}\beta + \beta^\top \mathbf{X}^\top \mathbf{X}\beta,$$

$$\frac{\partial \mathrm{RSS}(\beta)}{\partial \beta} = \mathbf{0}^\top - 2\mathbf{y}^\top \mathbf{X} + 2\beta^\top (\mathbf{X}^\top \mathbf{X}) = -2(\mathbf{y}^\top - \beta^\top \mathbf{X}^\top)\mathbf{X},$$

$$\frac{\partial \mathrm{RSS}(\beta)}{\partial \beta \partial \beta^\top} = \mathbf{X}^\top \mathbf{X} \succcurlyeq \mathbf{0}.$$

(a): $\|\mathbf{v}\|_2^2 = \langle \mathbf{v}, \mathbf{v} \rangle = \mathbf{v}^\top \mathbf{v}$, (b): multiplying out the terms, (c) $\frac{\partial \mathbf{c}^\top \mathbf{a}}{\partial \mathbf{a}} = \mathbf{c}^\top$, $\frac{\partial \mathbf{a}^\top \mathbf{B}\mathbf{a}}{\partial \mathbf{a}} = \mathbf{a}^\top (\mathbf{B} + \mathbf{B}^\top)$ [recipe].

# Solution – continued

- Positive semi-definiteness: $\mathbf{X}^\top \mathbf{X} \succcurlyeq \mathbf{0}$. Indeed, for any $\mathbf{v} \in \mathbb{R}^{p+1}$

$$\mathbf{v}^\top \mathbf{X}^\top \mathbf{X} \mathbf{v} = \langle \mathbf{X}\mathbf{v}, \mathbf{X}\mathbf{v} \rangle \geq 0.$$

- Normal equations [after taking $(\cdot)^\top$]:

$$-2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\beta) = \mathbf{0} \overset{/-2}{\Longleftrightarrow} \mathbf{X}^\top (\mathbf{y} - \mathbf{X}\beta) = \mathbf{0}.$$

  Intuition (recall cos), columns of $\mathbf{X} \perp \mathbf{y} - \mathbf{X}\beta$.

# Solution – continued

From the normal equations:

$$\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\beta) = \mathbf{0} \Leftrightarrow \mathbf{X}^\top\mathbf{X}\beta = \mathbf{X}^\top\mathbf{y}.$$

Assume that $\mathbf{X}$ has full column rank, i.e. $\exists \left(\mathbf{X}^\top\mathbf{X}\right)^{-1}$:

$$\hat{\beta} = \left(\mathbf{X}^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top\mathbf{y} \Rightarrow$$

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\beta} = \underbrace{\mathbf{X}\left(\mathbf{X}^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top}_{=:\mathbf{H}}\mathbf{y}.$$

- Recall: $\mathbf{X}^-$. $\mathbf{H}$ is called the hat matrix (puts a hat to $\mathbf{y}$).
- Prediction on a new point $\mathbf{x}_0$: $\hat{f}(\mathbf{x}_0) = (\tilde{\mathbf{x}}_0)^\top\hat{\beta}$.

# Solution: simpler form for $p = 1$

In case of $p = 1$:

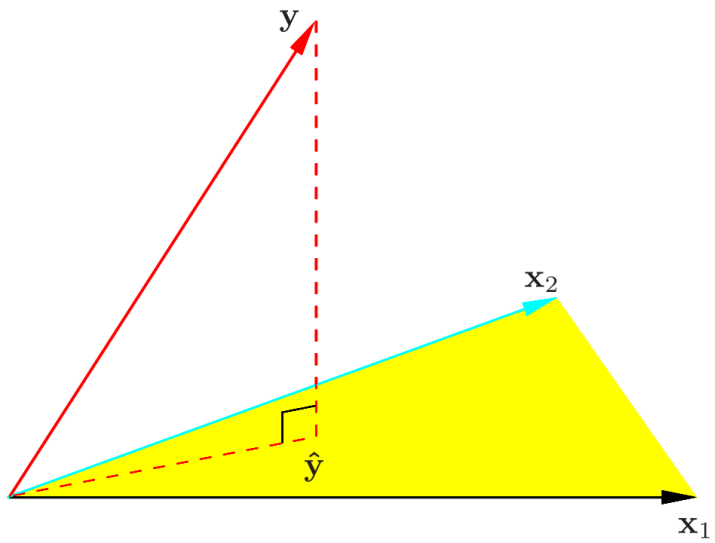
$$\bar{x} = \frac{1}{n} \sum_{i \in [n]} x_i, \qquad \bar{y} = \frac{1}{n} \sum_{i \in [n]} y_i,$$

the coefficients $\hat{\beta} = \left(\hat{\beta}_i\right)_{i=0}^{1} \in \mathbb{R}^2$ can be computed as

$$\hat{\beta}_1 = \frac{\sum_{i \in [n]} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i \in [n]} (x_i - \bar{x})^2},$$
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

# Solution: visually

Orthogonality: $\mathbf{X}^\top \left( \mathbf{y} - \underbrace{\mathbf{X}\hat{\beta}}_{\hat{\mathbf{y}}} \right) = \mathbf{0}$.




**H** is also called projection matrix.

**How to recognize an orthogonal projection?**

$\mathbf{P}^2 = \mathbf{P} = \mathbf{P}^\top$.

# The $\exists \left(\mathbf{X}^\top \mathbf{X}\right)^{-1}$ assumption

If $\nexists \left(\mathbf{X}^\top \mathbf{X}\right)^{-1}$:

- $\hat{\boldsymbol{\beta}}$ (coefficients): might not be unique.
- $\hat{\mathbf{y}}$ (projected $\mathbf{y}$): is still unique.
- In practice, typically regression softwares notice & handle it.



# Recall the model, fixed design

- Model (with $\mathbf{X}$ fixed), and coefficient estimate

$$\mathbf{y} = \mathbf{X}\beta + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I}); \qquad \hat{\beta} = \left(\mathbf{X}^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top\mathbf{y}.$$

- Distributional properties [$\mathbf{y} \sim \mathcal{N} \Rightarrow \hat{\beta} \sim \mathcal{N}$]: $\hat{\beta} \sim \mathcal{N}\left(\beta, \left(\mathbf{X}\mathbf{X}^\top\right)^{-1}\sigma^2\right)$.

  Indeed, by the affine transformation of normal variables

$$\mathbf{y} \sim \mathcal{N}(\mathbf{X}\beta, \sigma^2\mathbf{I}),$$
$$\hat{\beta} \sim \mathcal{N}(\mathbf{m}_\beta, \boldsymbol{\Sigma}_\beta), \text{ with}$$
$$\mathbf{m}_\beta = \left(\mathbf{X}^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top\mathbf{X}\beta = \beta,$$
$$\boldsymbol{\Sigma}_\beta = \left(\mathbf{X}^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top(\sigma^2\mathbf{I})\mathbf{X}\left(\mathbf{X}^\top\mathbf{X}\right)^{-1} = \left(\mathbf{X}\mathbf{X}^\top\right)^{-1}\sigma^2.$$

# Distributional properties

Variance:

- estimate

$$\hat{\sigma}^2 = \frac{1}{n-p-1} \sum_{i \in [n]} (y_i - \hat{y}_i)^2.$$

  Note: factor $\frac{1}{n-p-1}$ makes sure that $\mathbb{E}\left[\hat{\sigma}^2\right] = \sigma^2$ (unbiased).

- Distributional property:

$$(n-p-1)\hat{\sigma}^2 \sim \sigma^2 \chi^2_{n-p-1}.$$

Combining this, with $\hat{\boldsymbol{\beta}} \sim \mathcal{N}\left(\boldsymbol{\beta}, \left(\mathbf{X}\mathbf{X}^\top\right)^{-1} \sigma^2\right)$, and using the independence of $\hat{\sigma}^2$ and $\hat{\boldsymbol{\beta}}$, we get for $S \subseteq [p]$

1. test for $\boldsymbol{\beta}_S = \mathbf{0}$,
2. confidence interval for $\boldsymbol{\beta}_S$.

# Test $\beta_j = 0$, $j \in [p]$ fixed, i.e. $S := \{j\}$

- Aim:
  - $H_0$ (null hypothesis): $X_j$ has no effect on $Y$, i.e.

$$\beta_j = 0.$$

  - $H_1$ (alternative hypothesis): $X_j$ has effect on $Y$, i.e.,

$$\beta_j \neq 0.$$

- Test statistic: **Z-score** (a.k.a. **t-statistic**)

$$z_j = \frac{\hat{\beta}_j}{\hat{\sigma}\sqrt{v_j}}, \qquad v_j = \mathbf{e}_j^\top \operatorname{diag}\left(\left(\mathbf{X}^\top \mathbf{X}\right)^{-1}\right).$$

Nickname: $\mathrm{SE}(\hat{\beta}_j) = \hat{\sigma}\sqrt{v_j}$ (**standard error** associated with $\hat{\beta}_j$).

**Under $H_0$: $z_j \sim t_{n-p-1}$.**

# Test $\beta_j = 0$: given level $\alpha \in (0, 1)$, say $\alpha = 0.01, 0.05$

- Idea: we reject $H_0$ if $|z_j|$ is 'big'.
- In terms
  - quantile: Determine (read out from sheet) the quantile s.t.

$$P(|t_{n-p-1}| \leq q) = 1 - \alpha.$$

    Decision: If $|z_j| > q$, then reject $H_0$.
  - $p$-value:

$$\text{p-value} := P(|t_{n-p-1}| \geq z_j)$$

    Decision: if p-value $< \alpha$ (very unlikely observation), then reject $H_0$.

# Standard errors for $p = 1$

They take a simple form:

$$\mathrm{SE}^2\left(\hat{\beta}_0\right) = \hat{\sigma}^2\left[\frac{1}{n} + \frac{\bar{x}}{\sum_{i\in[n]}(x_i - \bar{x})^2}\right], \quad \mathrm{SE}^2\left(\hat{\beta}_1\right) = \frac{\hat{\sigma}^2}{\sum_{i\in[n]}(x_i - \bar{x})^2}.$$

Intuition:

- As $x_i$ is more spread out, the denominator of $\mathrm{SE}^2\left(\hat{\beta}_1\right)$ increases $\Rightarrow \mathrm{SE}^2\left(\hat{\beta}_1\right)$ decreases.

# Test $\beta_S = \mathbf{0}$; $S \subseteq [p]$ fixed; given level $\alpha \in (0, 1)$

- We have a
  - model with $p_1 + 1$ parameters.
  - smaller nested model $p_0 + 1$ parameters.

  i.e. $p_1 - p_0$ parameters constrained to be zero ($H_0$; aim: elimination).
- $F$-statistic:

$$F = \frac{(\mathrm{RSS}_0 - \mathrm{RSS}_1)/(p_1 - p_0)}{\mathrm{RSS}_1/(n - p_1 - 1)}$$

$\mathrm{RSS}_i := \mathrm{RSS}$ for the model with $p_i + 1$ parameters ($i \in \{0, 1\}$).

Under $H_0$ (smaller model is correct): $F \sim F_{p_1 - p_0, n - p_1 - 1}$.

- Notes:
  1. using $z_j \Leftrightarrow$ $F$-statistic for dropping one coefficient $\beta_j$.
  2. for large $n$, $F_{p_1 - p_0, n - p_1 - 1} \approx \chi^2_{p_1 - p_0}/(p_1 - p_0)$.

# Until now: $\beta \sim \mathcal{N} \Rightarrow$ test $\beta_S = \mathbf{0}$

- Recall:

$$\hat{\beta} \sim \mathcal{N}\left(\beta, \left(\mathbf{X}\mathbf{X}^\top\right)^{-1} \sigma^2\right).$$

- $\Rightarrow$ Now: confidence interval/set.

1. single coordinate: $1-\alpha$ confidence interval for $\beta_j$

$$\left(\hat{\beta}_j - z^{(1-\frac{\alpha}{2})} \mathrm{SE}\left(\hat{\beta}_j\right), \hat{\beta}_j + z^{(1-\frac{\alpha}{2})} \mathrm{SE}\left(\hat{\beta}_j\right)\right),$$

$z^{(1-a)} :=$ $1-a$ quantile of $\mathcal{N}(0,1)$. With

$$\alpha = 0.05, \qquad z^{(1-0.025)} = 1.96 \approx 2,$$
$$\alpha = 0.1, \qquad z^{(1-0.05)} = 1.645.$$

$\Rightarrow$ For an $\approx 95\%$ confidence interval, standard practice:

$$\left(\hat{\beta}_j - 2\mathrm{SE}\left(\hat{\beta}_j\right), \hat{\beta}_j + 2\mathrm{SE}\left(\hat{\beta}_j\right)\right).$$

# Confidence set: whole $\boldsymbol{\beta}$

Recall:

$$\hat{\boldsymbol{\beta}} \sim \mathcal{N}\left(\boldsymbol{\beta}, \left(\mathbf{X}\mathbf{X}^\top\right)^{-1}\sigma^2\right).$$

$\Rightarrow$ Confidence set for $\boldsymbol{\beta}$:

$$C_{\boldsymbol{\beta}} = \left\{\boldsymbol{\beta} \in \mathbb{R}^{p+1} \,:\, \left(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}\right)^\top \mathbf{X}^\top\mathbf{X}\left(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}\right) \leq \hat{\sigma}^2\left(\chi^2_{p+1}\right)^{(1-\alpha)}\right\},$$

where $\left(\chi^2_k\right)^{(1-\alpha)}$ is the $1-\alpha$ quantile of the distribution $\chi^2_k$.

**What kind of set is $C_{\boldsymbol{\beta}}$?**

# Confidence set: whole $\beta$ – continued

- Obtained confidence set:

$$C_\beta = \left\{\beta \in \mathbb{R}^{p+1} \,:\, \left(\beta - \hat{\beta}\right)^\top \underbrace{\mathbf{X}^\top \mathbf{X}}_{=:\mathbf{M}} \left(\beta - \hat{\beta}\right) \leq \underbrace{\hat{\sigma}^2 \left(\chi^2_{p+1}\right)^{(1-\alpha)}}_{=:r^2}\right\},$$

- Consider

$$\langle \mathbf{a}, \mathbf{b} \rangle_\mathbf{M} := \mathbf{a}^\top \mathbf{M} \mathbf{b}.$$

Notes:

- Classically, $\mathbf{M} = \mathbf{I}$.
- $\langle \cdot, \cdot \rangle_\mathbf{M}$ is an inner product if $\mathbf{M} \succ \mathbf{0}$ [i.e., $\langle \mathbf{a}, \mathbf{a} \rangle_\mathbf{M} \geq 0, = 0 \Leftrightarrow \mathbf{a} = \mathbf{0}$]
- One can take the induced norm, and ball of radius $r$ around zero:

$$\|\mathbf{a}\|_\mathbf{M} := \sqrt{\langle \mathbf{a}, \mathbf{a} \rangle_\mathbf{M}}, \qquad B_\mathbf{M}(\mathbf{0}) := \{\beta \,:\, \|\beta\|_M^2 \leq r^2\}$$

$\Rightarrow$ $C_\beta$ is a ball with center $\hat{\beta}$ (or in the classical sense an ellipsoid)

$$C_\beta = B_\mathbf{M}\left(\hat{\beta}\right) = \left\{\beta \,:\, \left\|\beta - \hat{\beta}\right\|_\mathbf{M}^2 \leq r^2\right\}.$$

# Once we checked $\beta \neq \mathbf{0}$

2 widely-used measures of fit:

1. $\mathrm{RSE}$ (recall):

$$\mathrm{RSE} = \sqrt{\frac{1}{n-p-1}\mathrm{RSS}}, \qquad \mathrm{RSS} = \sum_{i \in [n]} (y_i - \hat{y}_i)^2.$$

It measures the lack of fit to data in the units of $Y$.

# Once we checked $\beta \neq \mathbf{0}$ – continued

2. $\mathrm{R}^2$:

$$\mathrm{R}^2 = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}}, \qquad \mathrm{RSS} = \sum_{i \in [n]} (y_i - \hat{y}_i)^2, \qquad \mathrm{TSS} = \sum_{i \in [n]} (y_i - \bar{y})^2.$$

Intuition:

- RSS: unexplained variability in the response left after regression,
- TSS: total variability in the response,
- $\mathrm{R}^2$: proportion of variability in $Y$ explained by $X$; the higher the better.
- Value: $\mathrm{R}^2 \in [0, 1]$. Relation to empirical correlation ($p = 1$):

$$r = \widehat{\mathrm{corr}}(X, Y) = \frac{\sum_{i \in [n]} (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i \in [n]} (x_i - \bar{x})^2 \sum_{i \in [n]} (y_i - \bar{y})^2}}.$$

With this notation, $\mathrm{R}^2 = r^2$.

# Variants: qualitative predictors

- Aim:

  credit card balance $\approx \beta_0 + \beta_1 \times$ being the owner of a house.

- Dummy variables:

$$x_i = \begin{cases} 1 & \text{if the } i^{th} \text{ person owns a house} \\ 0 & \text{otherwise} \end{cases}$$

- Resulting model/observations:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i \in [n].$$

# Variants: interaction/synergy terms

- Standard model:

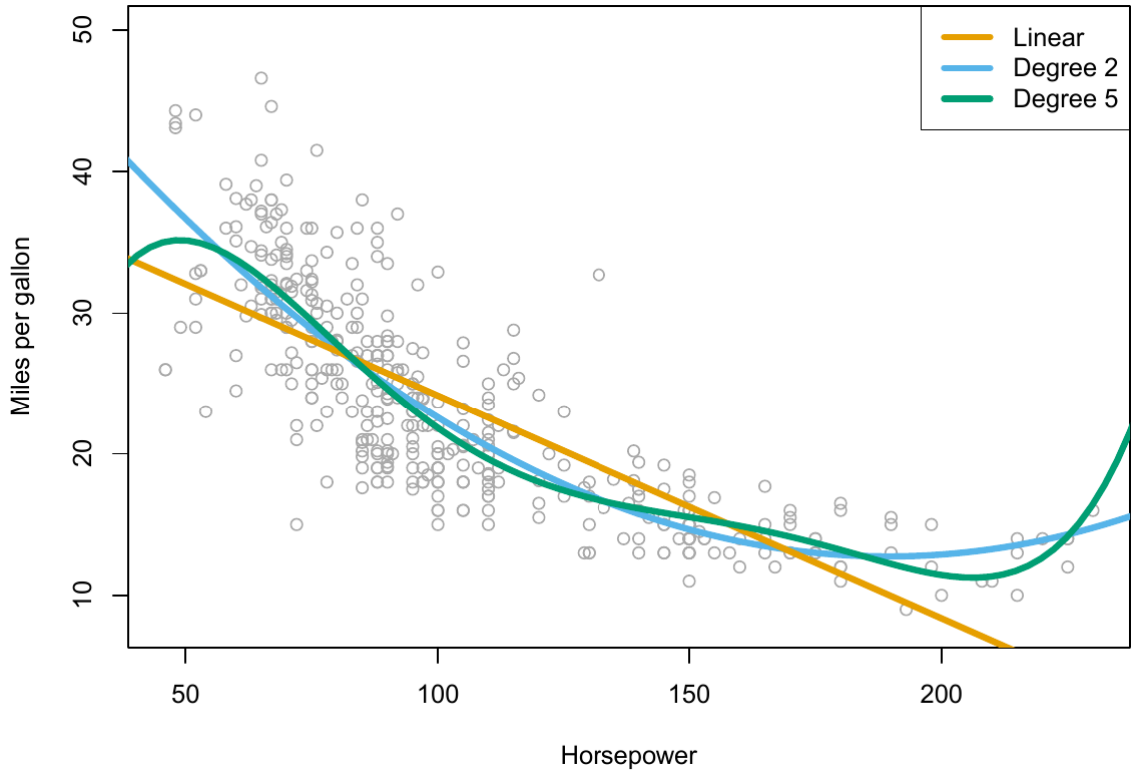
$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon.$$

- Model with interaction between $X_1$ and $X_2$:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \epsilon.$$

# Variants: polynomial features

- Example: mpg (gas mileage in miles per gallon) versus horsepower.




Instead of a linear fit (which looks off):

$$\text{mpg} = \beta_0 + \beta_1 \times \text{horsepower} + \beta_2 \times \text{horsepower}^2 = \langle \boldsymbol{\beta}, \varphi(X) \rangle,$$

$$\varphi(X) = \left(1, X, X^2\right).$$

# Linear regression vs kNN regression

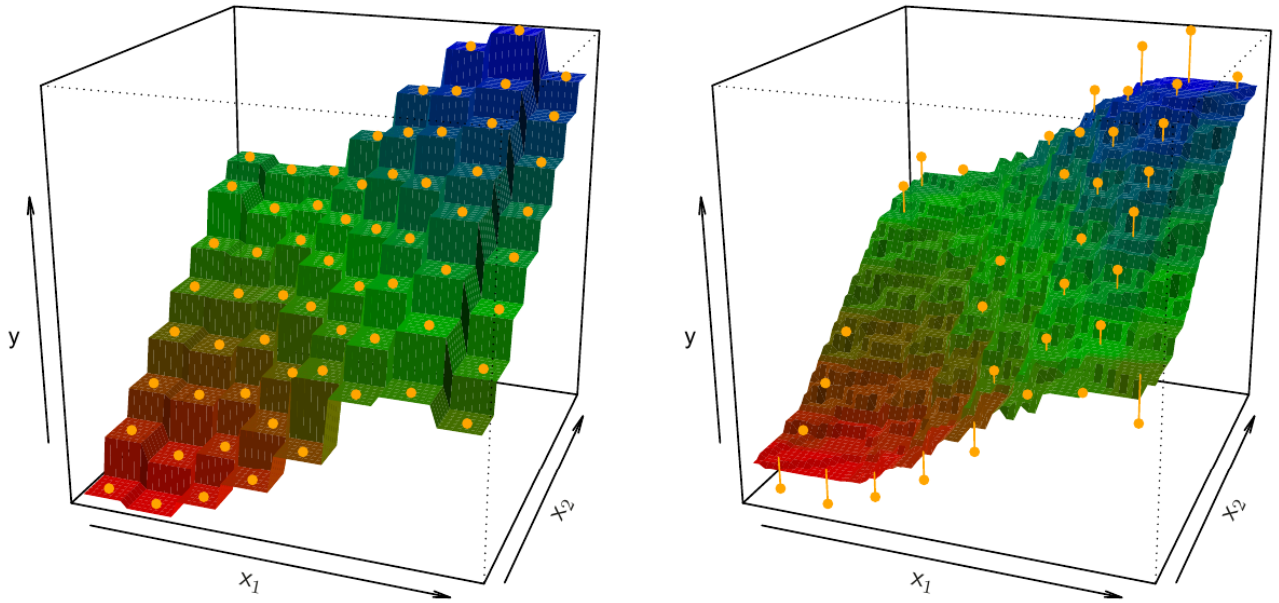
- Linear regression:
  - parametric technique,
  - easy to fit & interpret, tests,
  - strong assumptions.
- KNN regression (= regression variant of KNN classifier):
  - nonparametric approach,
  - given: $K \in \mathbb{Z}^+$, training data, $\mathbf{x}_0$ test point,
  - $\mathcal{N}_0 = K$ training observations closest to $\mathbf{x}_0$.

$$\hat{f}(\mathbf{x}_0) = \frac{1}{K} \sum_{\mathbf{x}_i \in \mathcal{N}_0} y_i.$$

It takes the average prediction of the KNNs.

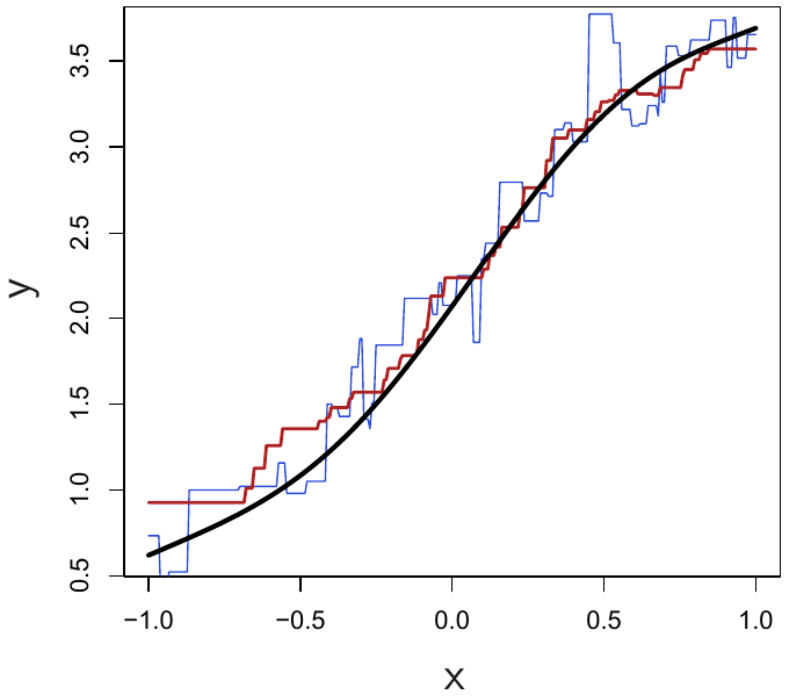
# KNN regression: example with $p = 2$

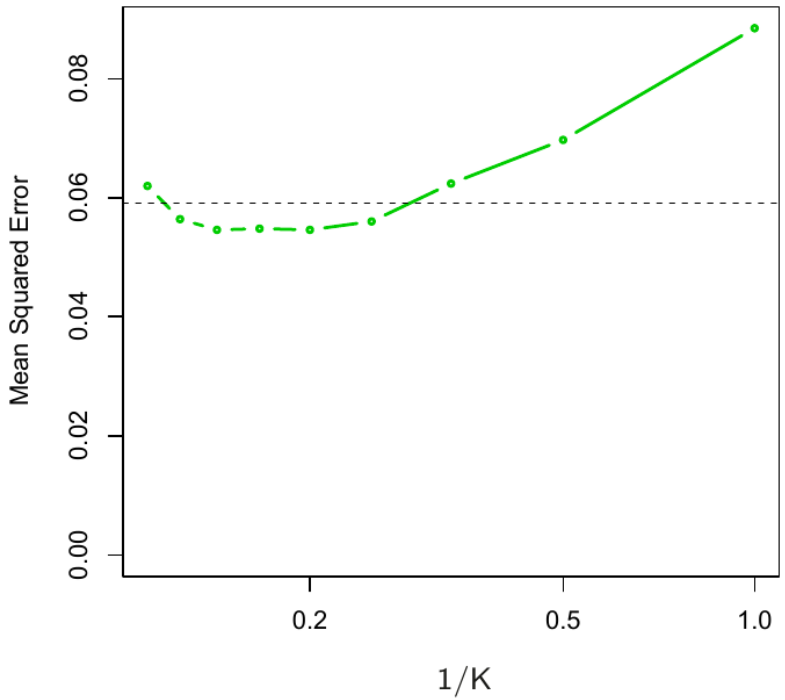
Left: $K = 1$ (rougher), right: $K = 9$ (smoother).



# Comparison of linear and KNN regression-1

- setting: $\approx$ linear relation.
- left: fit with $K = 1$ (blue) and $K = 9$ (red); right: test MSE vs different $K$-s, with linear regression baseline (black).

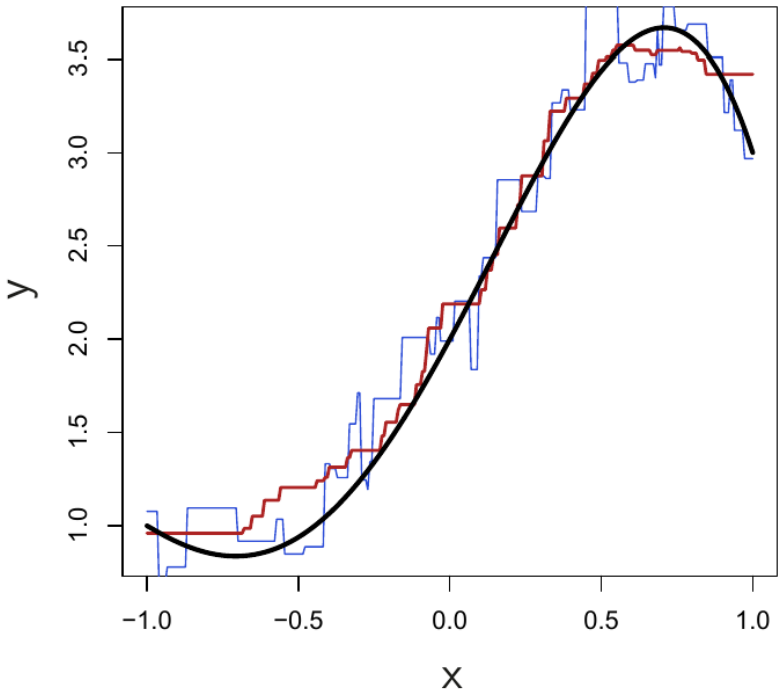


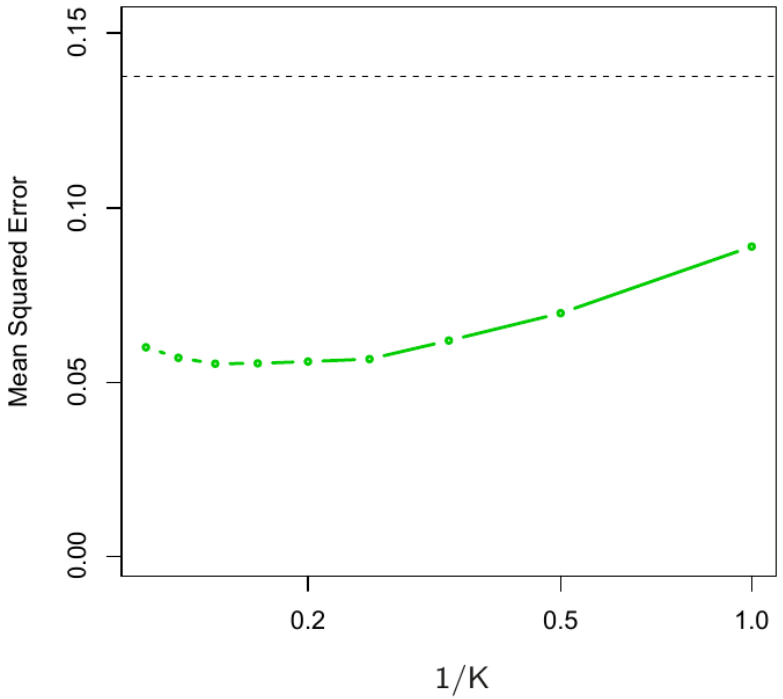




$\Rightarrow$ KNN regression: better for $K \geq 4$ (l.h.s. on the r.h.s.).

# Comparison of linear and KNN regression-2

- setting: increased non-linearity.
- left: fit with $K = 1$ (blue) and $K = 9$ (red); right: test MSE vs different $K$-s, with linear regression baseline (black).




$\Rightarrow$ KNN regression: better overall; in higher dimension it can struggle (curse of dimensionality).

# Local summary

- probability recap; key distributions: $\mathcal{N}$, $\chi^2$, $t$, $F$,
- least squared problem: formulation, solution, projection view,
- testing, confidence intervals,
- measures of fit,
- linear regression tweaks, and towards nonlinearity.

